

Gig Hiring Bench

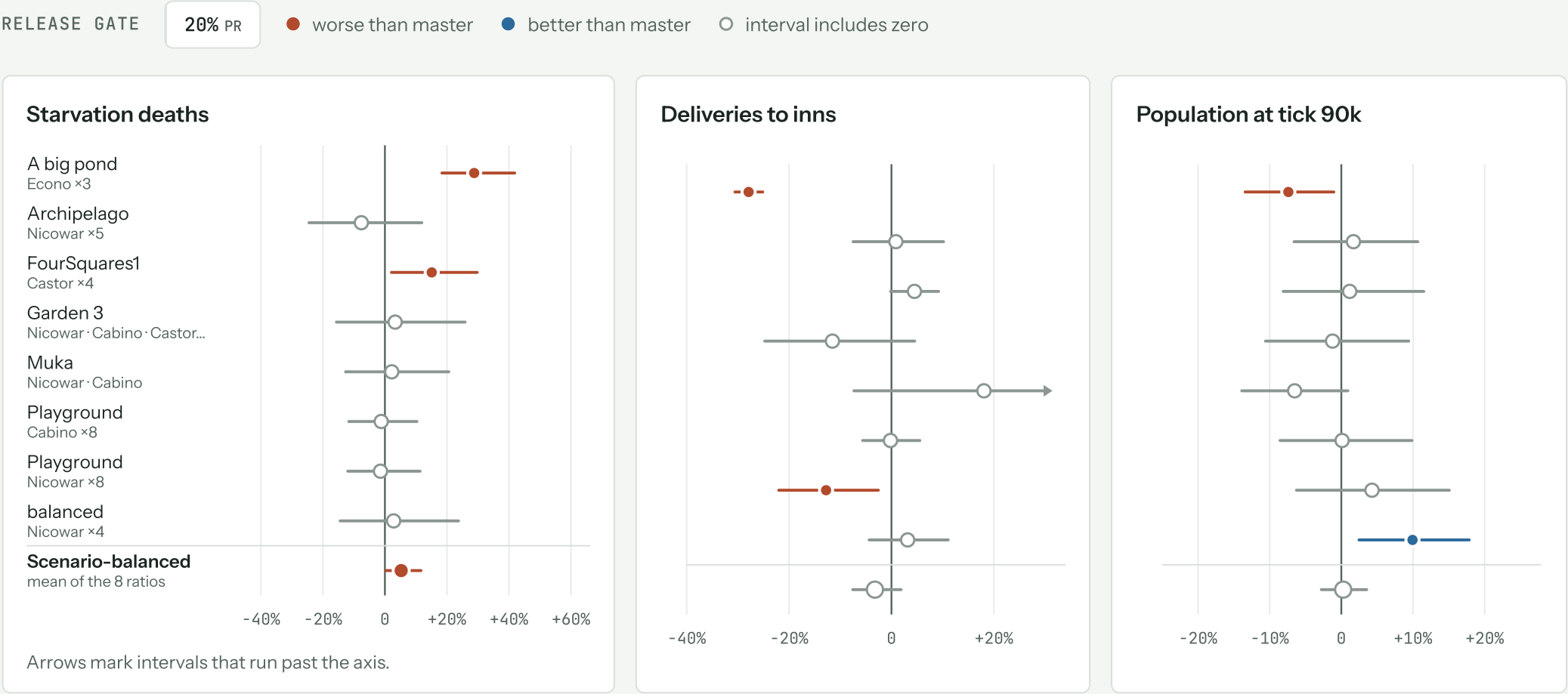
Releasing a worker after each delivery does **not** reduce starvation. Across eight AI scenarios it is mostly indistinguishable from master. With Econo, which keeps nearly half its workers idle, it cuts food deliveries to inns by **28%** and raises starvation deaths by **29%**.

What the numbers say

ECONO · A BIG POND · PR'S 20% GATE +28.8% more starvation deaths by tick 90,000. Inn deliveries fall 27.9% and population falls 7.4%. 95% CI +18.4% to +41.9% · n = 60	GISZMO'S WINDOW, RE-MEASURED +5.6% starvation deaths on Playground, 8× Nicowar, ticks 24,576–40,960. The PR reported -30%, which lies outside this interval. 95% CI -12.0% to +27.1% · n = 60	SCENARIO-BALANCED · 20% GATE -0.1% deliveries by tick 90,000, averaged over the eight scenarios. Population at tick 90,000 changes +0.3%, with a CI of -2.8% to +3.5%. 95% CI -2.9% to +2.9%	FOOD REACHING INNS · POOLED -6.5% inn deliveries at the 20% gate, and below master at every threshold, including 50%. This is the most consistent signal. 95% CI -9.6% to -3.4% · n = 480
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Effect by scenario

Change against master on the same map, AIs and seed, at tick 90,000 (or the last tick, if the game ended earlier). Each dot is the ratio of means, and the whisker is the 95% bootstrap interval over 60 seeds. A filled dot means the interval excludes zero.

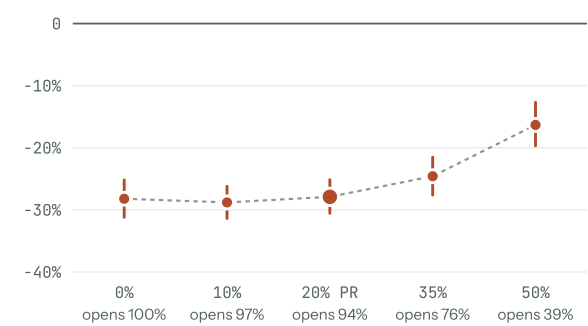


At the 20% gate, workers are released after 51% of deliveries across all games, ranging from 16% to 94% depending on the scenario.

Econo: the harm scales with how often the gate opens

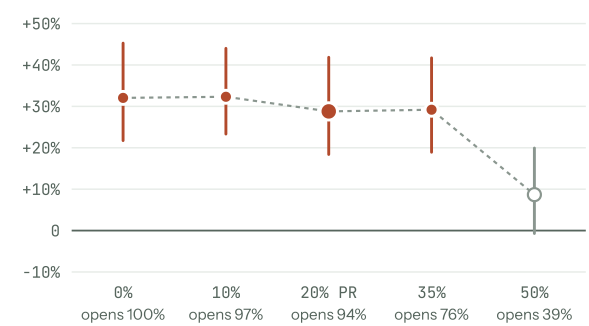
A big pond, 3× Econo. Econo keeps about 46% of its workers idle, so a low threshold opens the gate on almost every delivery. As the threshold rises and the gate opens less often, the loss of food deliveries and the extra deaths shrink. That dose response points to a real mechanism rather than noise.

Deliveries to inns



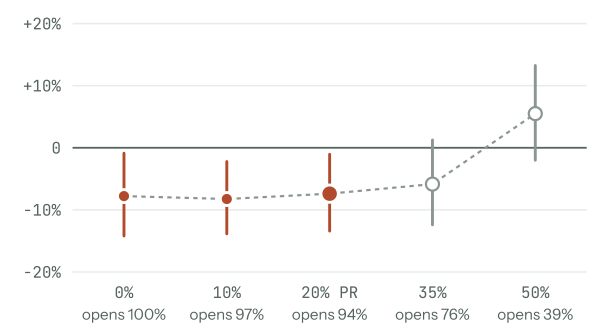
Release gate: the share of a team's workers that must be idle before a delivery releases its worker.

Starvation deaths



Release gate: the share of a team's workers that must be idle before a delivery releases its worker.

Population at tick 90k



Release gate: the share of a team's workers that must be idle before a delivery releases its worker.

Read single-scenario swings with care

Any change to hiring reshuffles a whole game: the same seed ends with the same winner only 15–93% of the time, depending on the scenario. On Playground with 8× Nicowar, every threshold makes games end in a win more often (67% on master, 78–88% with the PR). That includes the 50% gate, which releases on only 2% of deliveries. All five thresholds share one master baseline, and this page shows dozens of intervals, so expect a few ±10% results that are really noise. The Econo result survives both concerns: it is large, holds at every threshold, and shrinks as the gate opens less.

All scenarios

Master mean → PR mean, then change and 95% CI, for the gate selected above. "Gate opens" is the share of post-delivery decisions that released the worker. "Games won" is the share of games that ended with a winner before tick 90,000.

SCENARIO	DELIVERIES	INN DELIVERIES	STARVED	POP. 60K	POP. 90K	GATE OPENS	GAMES WON
A big pond Econo ×3	22,666 → 21,679 -4.3% [-6.3%, -2.3%]	7,452 → 5,373 -27.9% [-30.6%, -25.1%]	871.1 → 1,122 +28.8% [+18.4%, +41.9%]	1,057 → 766.2 -27.5% [-30.7%, -24.2%]	779.5 → 721.9 -7.4% [-13.4%, -1.0%]	94%	100% → 100%
Archipelago Nicowar ×5	10,621 → 10,703 +0.8% [-5.6%, +8.0%]	3,532 → 3,562 +0.9% [-7.5%, +10.2%]	21.5 → 19.9 -7.6% [-24.5%, +11.9%]	371.2 → 378.9 +2.1% [-4.3%, +8.7%]	381.3 → 387.8 +1.7% [-6.6%, +10.7%]	47%	53% → 48%
FourSquares1 Castor ×4	11,365 → 11,853 +4.3% [+0.1%, +8.5%]	6,350 → 6,635 +4.5% [-0.2%, +9.2%]	60.5 → 69.6 +15.1% [+2.1%, +29.8%]	267.6 → 297.2 +11.1% [+3.3%, +19.6%]	328.3 → 332.1 +1.2% [-8.1%, +11.5%]	84%	0% → 0%
Garden 3 Nicowar · Cabino · Castor · Warrush	6,356 → 5,856 -7.9% [-18.6%, +4.5%]	2,128 → 1,883 -11.5% [-24.8%, +4.5%]	155 → 160.1 +3.3% [-15.7%, +25.9%]	184.2 → 176.9 -3.9% [-12.9%, +6.5%]	179.8 → 177.6 -1.2% [-10.6%, +9.4%]	33%	83% → 90%
Muka Nicowar · Cabino	2,113 → 2,361 +11.7% [-3.2%, +28.7%]	713.7 → 842.5 +18.1% [-7.3%, +52.1%]	24.2 → 24.7 +2.2% [-12.7%, +20.6%]	107.7 → 103.3 -4.1% [-10.6%, +3.2%]	106 → 99.1 -6.5% [-13.9%, +0.9%]	19%	98% → 97%
Playground Cabino ×8	8,227 → 8,178 -0.6% [-4.5%, +3.4%]	3,181 → 3,175 -0.2% [-5.6%, +5.5%]	273.3 → 270.1 -1.2% [-11.7%, +10.3%]	126.2 → 126.1 -0.1% [-8.1%, +8.1%]	92.4 → 92.5 +0.1% [-8.6%, +9.9%]	33%	65% → 60%
Playground Nicowar ×8	11,248 → 10,537 -6.3% [-13.1%, +1.3%]	4,861 → 4,240 -12.8% [-22.0%, -2.5%]	165 → 162.6 -1.4% [-12.0%, +11.3%]	191 → 178.7 -6.5% [-14.2%, +2.1%]	153.6 → 160.2 +4.3% [-6.3%, +15.1%]	16%	67% → 85%
balanced Nicowar ×4	19,794 → 20,139 +1.7% [-4.5%, +8.3%]	6,848 → 7,064 +3.2% [-4.4%, +11.1%]	118.6 → 122 +2.8% [-14.4%, +23.7%]	425.9 → 438.8 +3.0% [-3.9%, +9.7%]	356.7 → 392.1 +9.9% [+2.5%, +17.8%]	23%	25% → 22%
All 8 scenarios pooled	11,549 → 11,413 -1.2% [-3.3%, +0.9%]	4,383 → 4,097 -6.5% [-9.6%, -3.4%]	211.1 → 243.9 +15.5% [+9.0%, +22.5%]	341.3 → 308.3 -9.7% [-12.7%, -6.6%]	297.2 → 295.4 -0.6% [-4.0%, +2.9%]	51%	62% → 63%

How this was measured

Builds

- **master**: origin/master at 63cefebbc with read-only telemetry.
- **PR**: #254 rebased onto the same master (VERSION_MINOR 100, CORN→WHEAT), with the same telemetry. The gate threshold is set by GLOB2_GIG_RELEASE_PERCENT: 0 always releases (the first commit), 20 is the PR, 101 never releases.
- Telemetry prints per-team deliveries, inn and site deliveries, starvation vs other deaths, idle workers, population and release/keep counts every 1,024 ticks. The simulation never reads it.

Limits

- AI players only. Nobody has played the PR, and how it feels to play is still an open question for review.
- Maxima and Cortex weren't in the matrix: the master base predates Maxima (#264), and Cortex wasn't included.
- No scenario with scarce workers (few workers, several hungry inns) was built, which is where the PR expects its gain.
- No placebo build (a harmless change that only reshuffles games) was run, so the noise floor is inferred from the high thresholds.

Checks before the batch

- Plain master and master with telemetry give identical per-team timelines on a mixed 4-AI Garden 3 game.
- The PR build at 101% matches master on every telemetry column, so all differences come from the releases themselves.
- Reruns are deterministic, and results match across machines: gcc 15 (therig, pharaoh-dev-2/3) and gcc 13 (devlaptop) produce identical output.
- pharaoh-dev-1 was excluded: its compiler segfaulted repeatedly on different files, which points to hardware trouble.

Design

- 8 scenarios × 60 seeds × 6 builds = 2,880 games, each from tick 0 to 90,000 unless someone wins first. None crashed or failed.
- Every build for a given seed ran on the same machine.
- Scenario tables use the ratio of means with a paired bootstrap (4,000 resamples of games). "Scenario-balanced" averages the eight per-scenario ratios, resampling within each scenario, so Econo's large starvation counts don't dominate.

```
~/gigbench/{master,pr}      remote trees (therig, devlaptop, pharaoh-dev-2/3)
scratchpad/gigbench/        run_job.sh · make_jobs.py · collect.sh · analyze.py
scratchpad/gigbench/results/ <map>/<matchup>/<seed>/<build>.log.gz  (2,880 logs)

GLOB2_TEST_SEED=<seed> GLOB2_GIG_BENCH=1024 GLOB2_GIG_RELEASE_PERCENT=20 \
  build/src/glob2 -test-games-nox 1 --map Playground --matchup nicowar,nicowar,...
```

Bench branches bench/gig-master and bench/gig-pr are local scratch worktrees and are not for merging. The telemetry patch is bench-only.